Learning Resources

Foundational Python + Tools

1. Python for Data Science (Beginner Friendly)

Teaches Python basics and tools needed for the whole project.

📌 https://www.datacamp.com/courses/intro-to-python-for-data-science
📌 https://realpython.com/python-data-science/

📊 Data Cleaning & Preprocessing

2. Data Wrangling with Pandas

Pandas is essential for cleaning datasets like the churn dataset.

- Handling missing values
- Encoding categorical variables
- Normalization / scaling

📌 Official Tutorial: https://pandas.pydata.org/docs/getting_started/intro_tutorials/index.html
📌 Practical guide: https://realpython.com/pandas-python-explore-dataset/

📉 Feature Engineering & Dimensionality

3. Feature Scaling & Encoding

How to scale data and convert categories to numbers — crucial for SVM/Decision Trees.

📌 scikit-learn preprocessing guide:
https://scikit-learn.org/stable/modules/preprocessing.html

📌 One-Hot Encoding tutorial:
https://www.analyticsvidhya.com/blog/2020/03/encoding-techniques-categorical-data/

📈 Supervised Learning (Classification)

4. Decision Trees

Simple and interpretable classification.

📌 scikit-learn tutorial:
https://scikit-learn.org/stable/modules/tree.html

📌 A clear example with code:
https://towardsdatascience.com/decision-trees-in-python-with-scikit-learn-a4da71e7d17a

5. Support Vector Machines (SVM)

Good for classification tasks like churn prediction.

📌 scikit-learn SVM guide:
https://scikit-learn.org/stable/modules/svm.html

📌 Intuitive SVM explanation:
https://www.datacamp.com/community/tutorials/svm-classification-scikit-learn-python

📊 Clustering (Unsupervised Learning)

6. K-Means Clustering

Students will use this to find customer groups.

📌 scikit-learn tutorial:
https://scikit-learn.org/stable/modules/clustering.html

📌 Beginner tutorial with visuals:
https://www.datacamp.com/community/tutorials/k-means-clustering-python

📉 Model Evaluation

7. Classification Metrics

How to calculate accuracy, precision, recall, F1 and confusion matrix.

📌 scikit-learn metrics:
https://scikit-learn.org/stable/modules/model_evaluation.html

📌 Visual explanations:
https://www.dataschool.io/simple-guide-to-confusion-matrix-terminology/

🧪 Train/Test Splitting & Sampling

8. Train/Test Split

Fundamental for evaluation.

📌 scikit-learn train/test:
https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.train_test_split.html